

- 1 (a) Fig. 1.1 is an electron micrograph showing a cross-section of a myelinated neurone.



Fig. 1.1

Name **A** and **B**.

A

B [2]

- (b) Explain what is meant by saltatory conduction **and** describe its effect on the transmission of a nerve impulse.

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..... [3]

- (c) A type of sea snail, *Conus purpurascens*, produces a toxin that blocks calcium ion channels in the presynaptic knob of a cholinergic synapse. The presence of this toxin results in no action potentials in the postsynaptic neurone.

Explain why the presence of this toxin results in **no** action potentials in the postsynaptic neurone.

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